

MOI HIGH-KABARAK 2025 PREDICTION



Kenya Certificate of Secondary Education

KCSE PREDICTION SERIES 1



448/2

Electricity

Paper 2

Time: 2 Hours

NAME.....

SCHOOL..... SIGN.....

INDEX NO..... ADM NO.....

INSTRUCTIONS TO CANDIDATES

- Write your **Name**, **Class** and **Index Number** in spaces provided
- There are **five** exercises in this examination
- Candidates are allowed **30** minutes for each exercise
- Each exercise will be awarded a maximum of **20 marks**
- At each station, candidates are **not allowed** to either view the previous station's work or read instructions for the other stations
- Attempt all the **five** exercises
- All dimensions are in **millimeters** unless otherwise stated
- Candidates should answer the questions in **English**

FOR EXAMINER'S USE ONLY

Exercise	1	2	3	4	5	TOTAL
MARKS(EC)						
MARKS(MC)						
GRAND TOTAL						

EXERCISE 1

Using the components, materials and equipment provided, carry out the following tasks

- a. Connect the circuit as shown in **Figure 1**. Let the examiner check your work (7 marks)

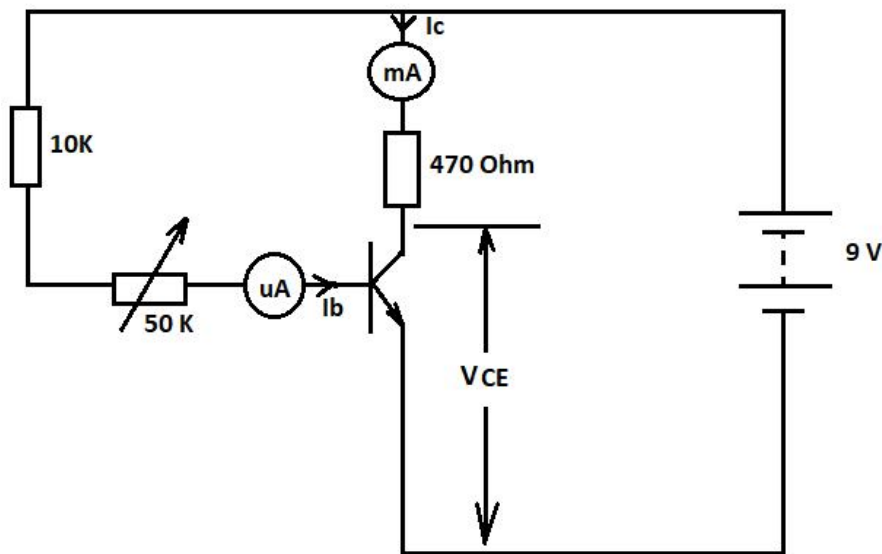


Figure 1

- b. Vary the base resistance R_B to obtain each of the base current I_B values shown in table 1. For each value of I_B , measure and record in the table the corresponding values of V_{CE} and I_C .

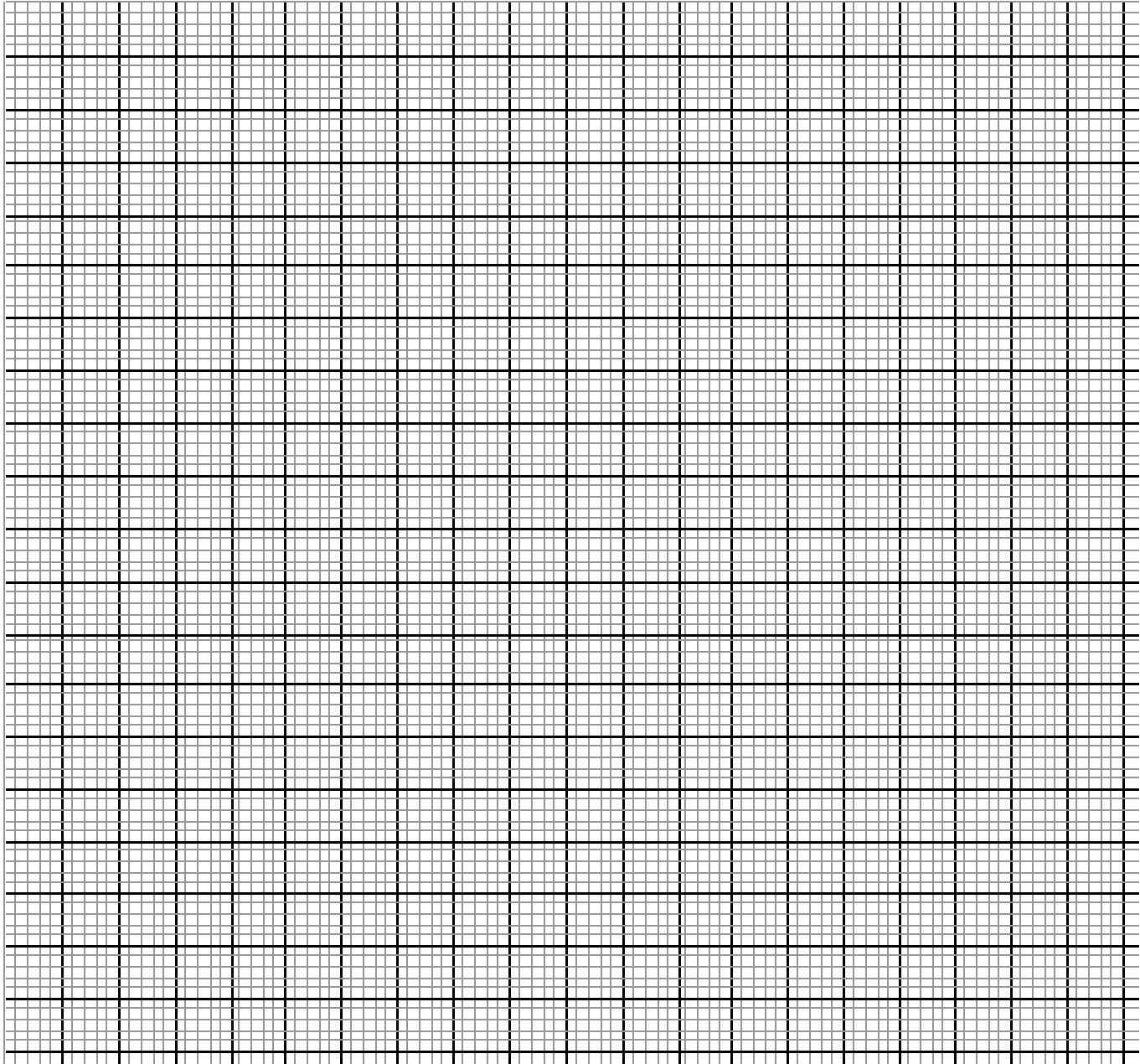
Table 1

$I_B(\mu A)$	$V_{CE}(V)$	$I_C(mA)$
200		
220		
240		
260		
280		
300		

- c. Using the values in table 1:

- i. Determine the current gain β when $I_B=240\mu A$ (2 marks)

ii. Plot a graph of I_C against V_{CE} (5 marks)



EXERCISE 2

Using the tools and equipment provided, make the model of a fluorescent lamp rain guard as shown in **Figure 2**

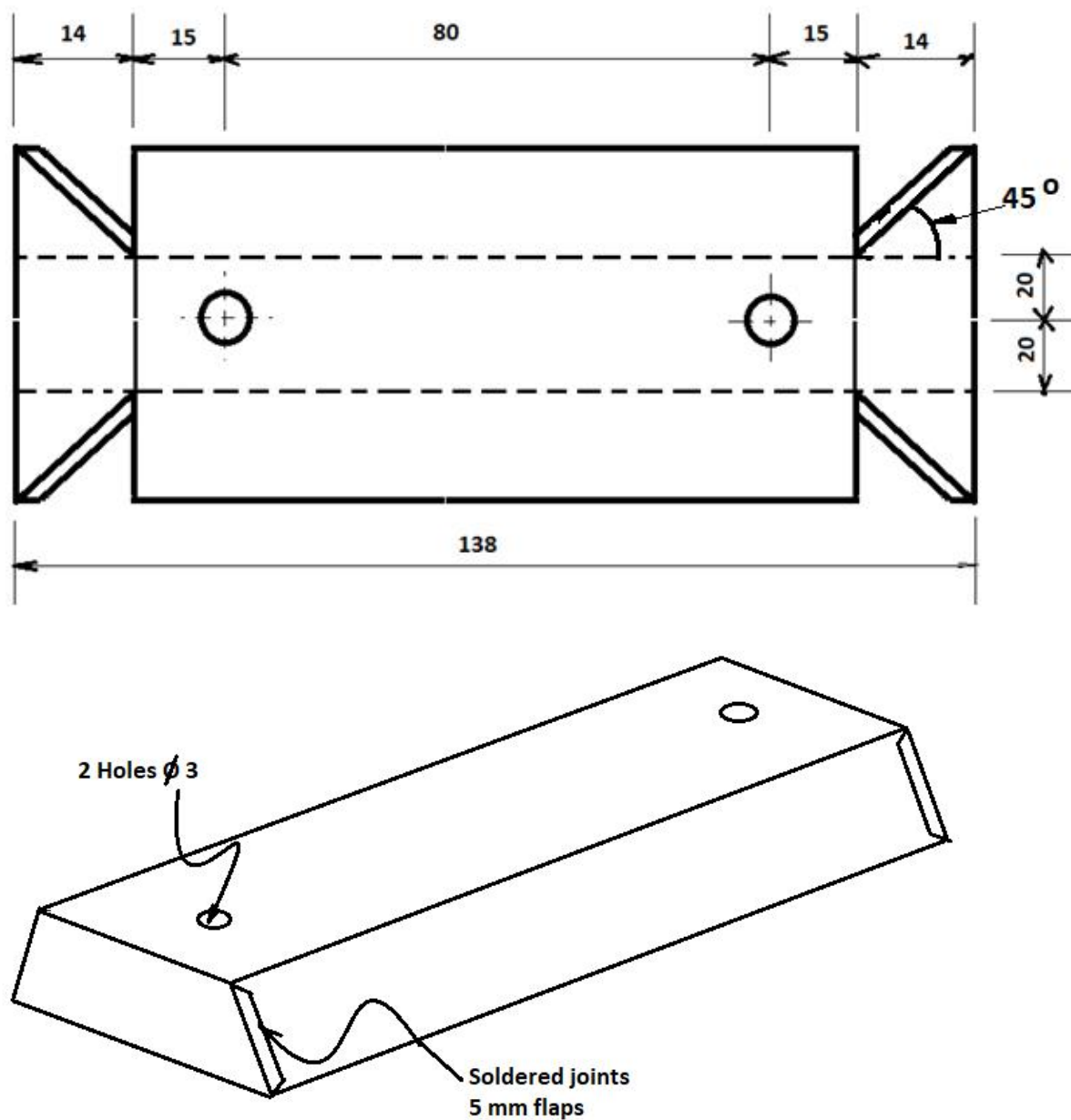


Figure 2

EXERCISE 3

Using the components, materials and equipment provided, perform the following tasks.

- a. Connect the circuit shown in **Figure 3** and let the examiner check your work (6 marks)

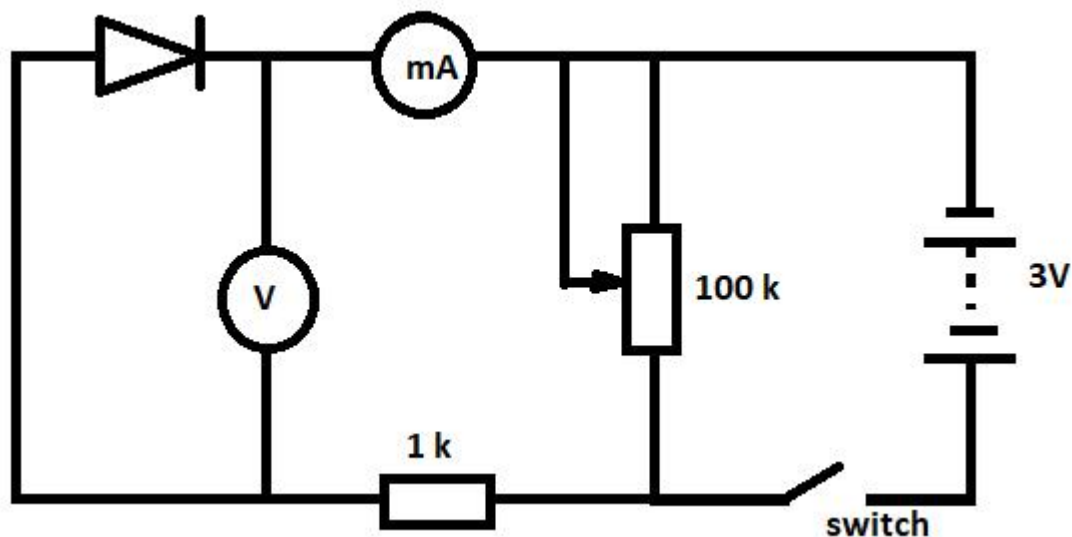


Figure 3

- b. Close the switch. Adjust the potentiometer to obtain voltage reading shown in **Table 2**. For each voltage reading obtained, measure and record the corresponding current values. (5 marks)

Table 2

Voltage (V)	0	0.5	0.7	0.9	1.3	1.7
Current (mA)						

- c. Plot a graph of current against voltage (6 marks)
- d. From the graph:
- Determine the resistance when the voltage is 1.3V
 - Give a reason for the shape of the graph (3 marks)

EXERCISE 4

Using the pre-fabricated circuit provided in **Figure 4**, perform the following tasks:

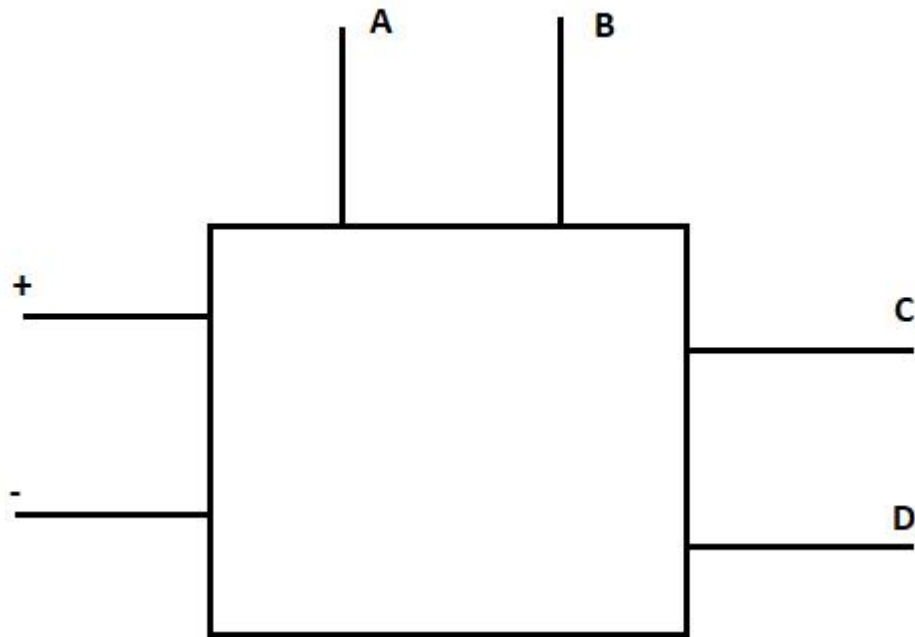


Figure 4

- a. With the switch open:
 - i. Connect Voltmeter between A and B (Observe polarity)
 - ii. Connect Voltmeter between C and D (Observe polarity) **(2 marks)**

(Let the examiner check your work)

- b. Close the switch. Adjust input voltage and complete the table below **(10 marks)**

Table 3

Input Voltage V_{in} (V)	0	1.0	2.0	3.0	4.0	5.0	6.0	7.0	8.0	9.0	10.0
Output voltage V_z (V)											

- c. Plot the graph of V_z against V_{in} **(5 marks)**

EXERCISE 5

Using PVC sheathed wiring, complete the installation shown in **Figure 5**, such that the bell is operated from either A or B
(20 marks)

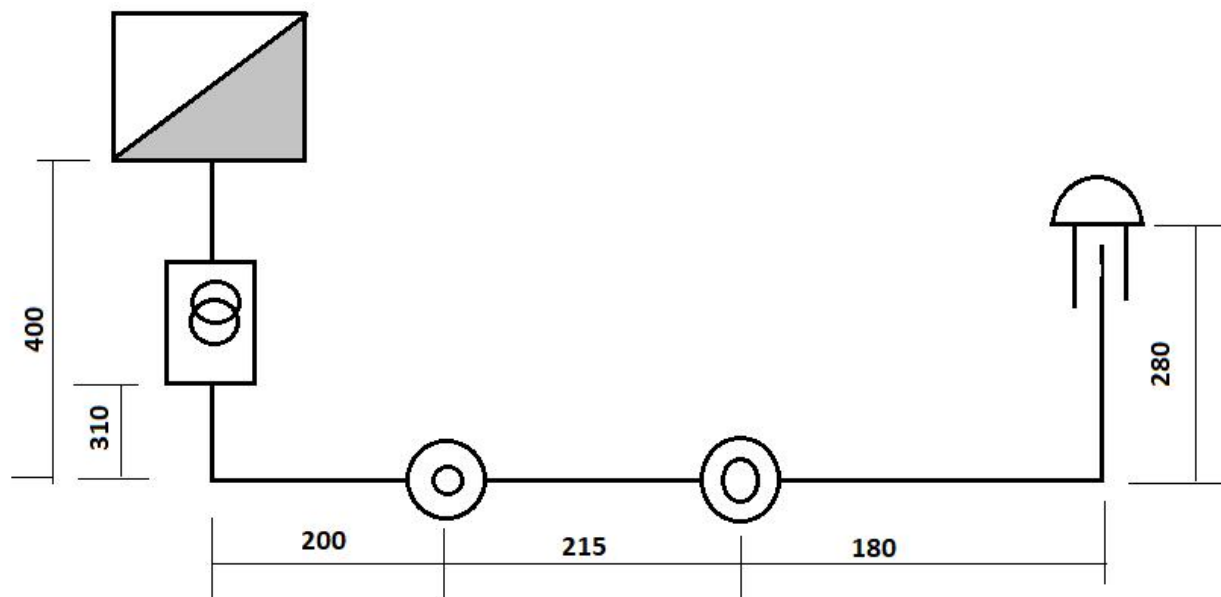


Figure 5